

Dave Andison's fire data preliminary analyses

Ceres Barros

March 28, 2018

NOTES - meeting with Dave Dec 6th

Carefull with no. of variables and sample size Weather variables: how to look at it? Weather at the begining, middle, end? Only averages like Colin did? Averages, variation?

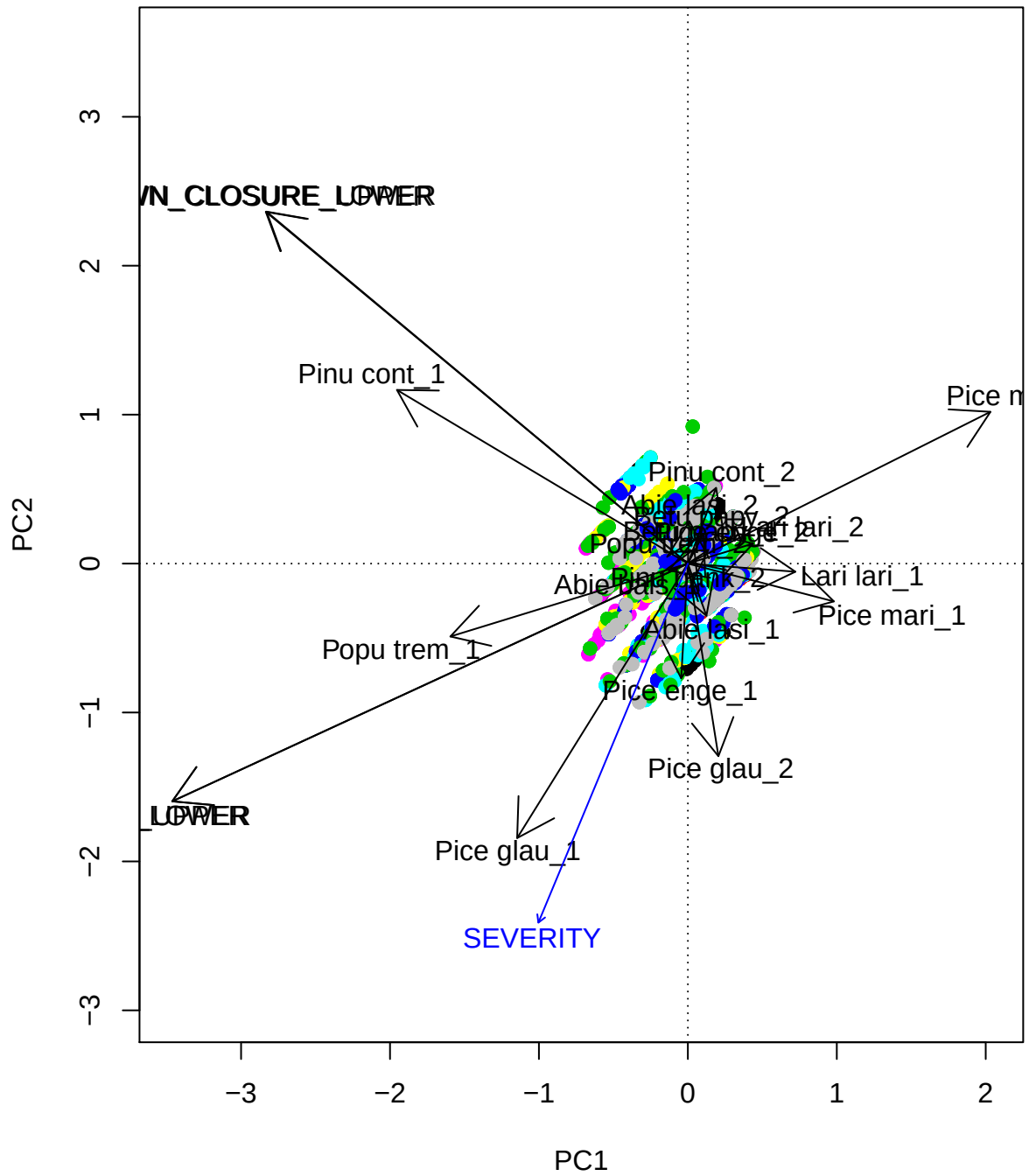
COMMUNITY COMPOSITION/STRUCTURE PCAs

PCA of dominant species percentage cover and height across all fire event patch types.

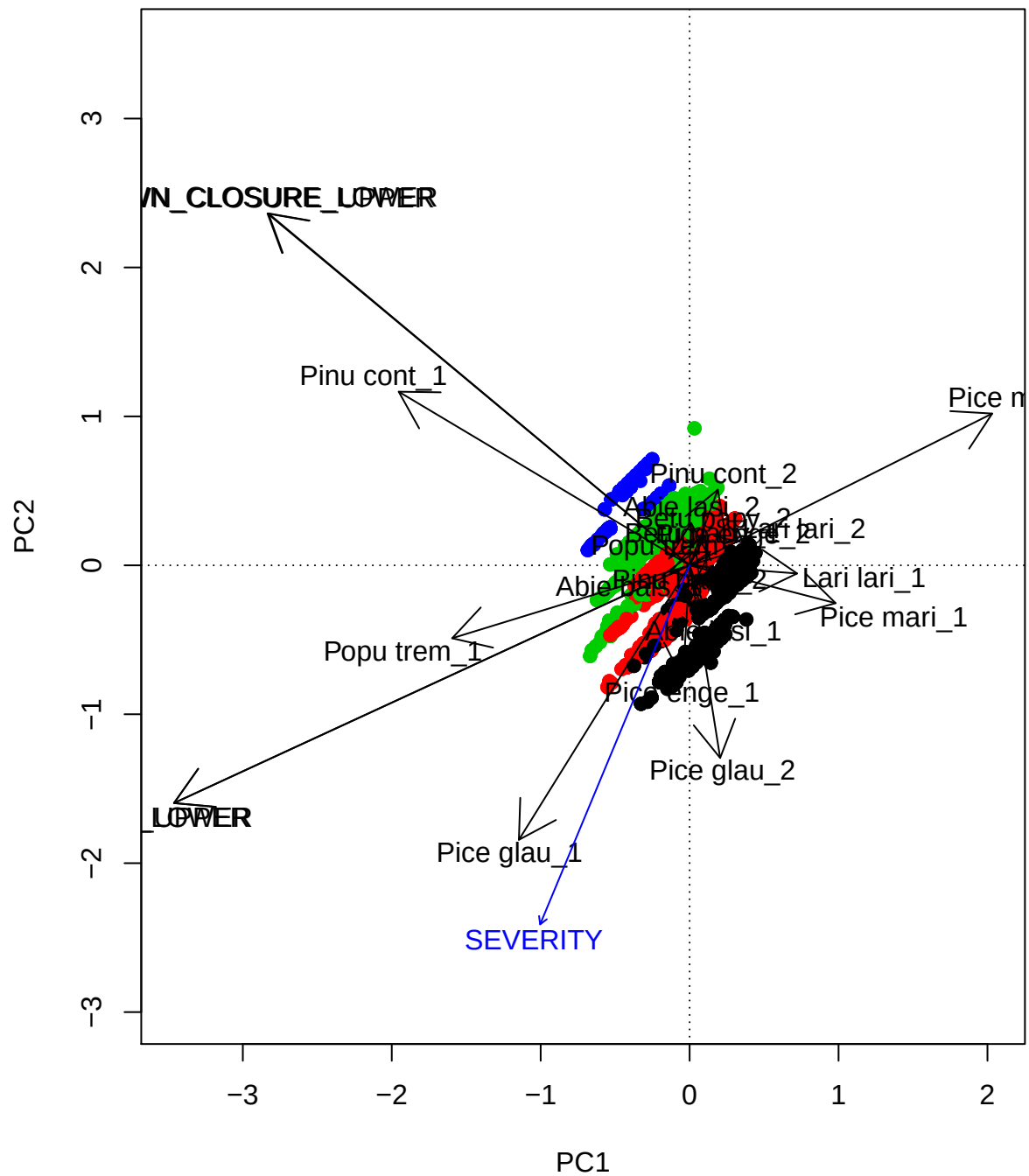
Blue arrow is severity in a continuous scale, fitted to the ordination *a posteriori*, using `env.fit`. Severity is higher in “plots” dominated by White spruce (*Picea glauca*), Engelman's spruce (*P. engelmannii*), Aspen (*Populus tremuloides*) and where trees are taller. It seems that canopy and understory levels usually highly correlated within species.

Also the points pattern doesn't seem related to fire identity, but to levels of crown closure.

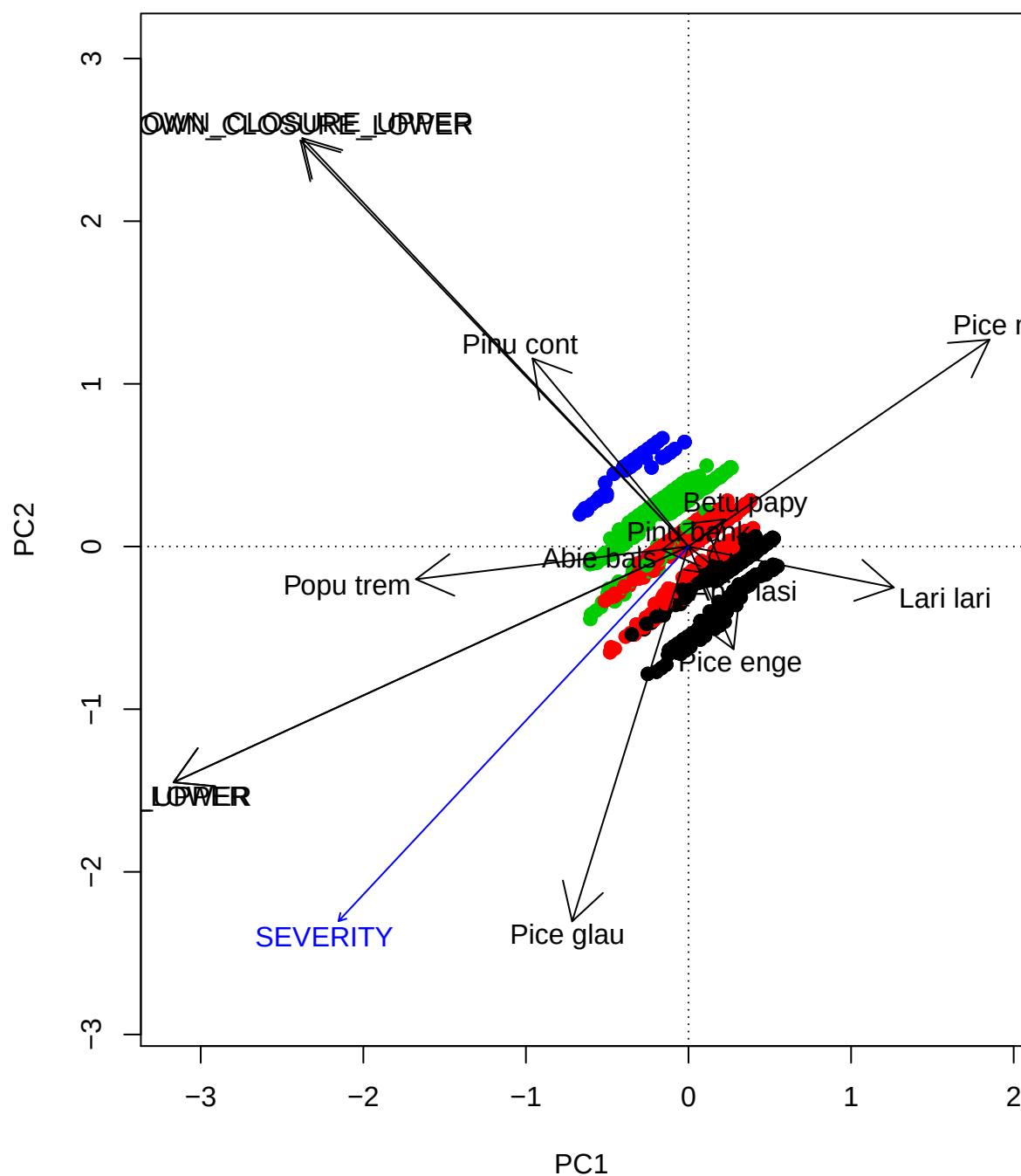
Coloured by fire name



Coloured by crown closure



Coloured by crown closure

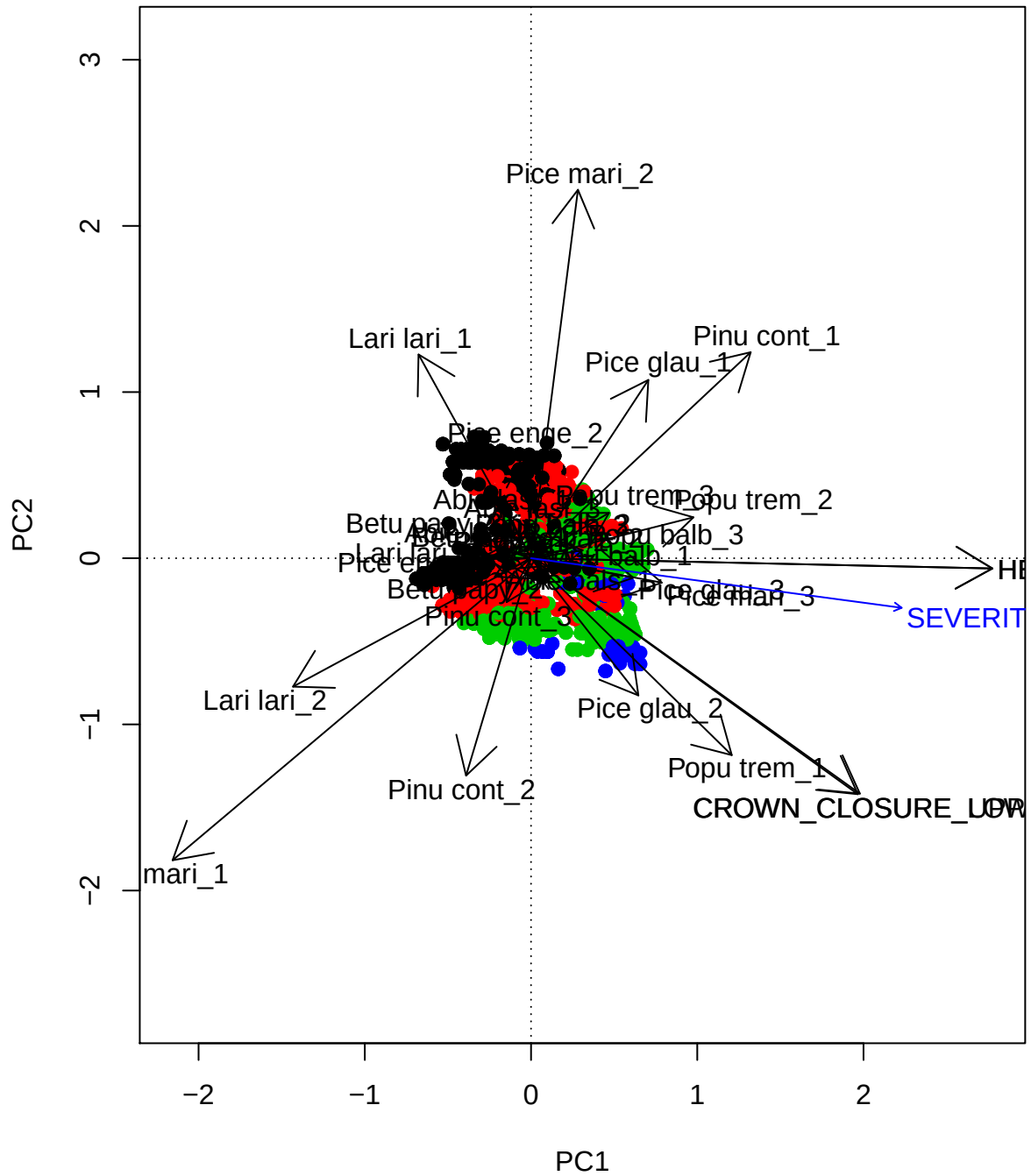


PCA with canopy and understory levels collapsed for tree species, but keeping dominant and subdominant spp separate (summed and rescaled to 0-1 - maybe not the best way)

Results are similar, but we now see some vegetation associations. Unlike before, crown closure now appears more correlated with severity. It seems higher severity levels coincide with forests where Aspen is the dominant

species and White spruce is the subdominant.

Coloured by crown closure

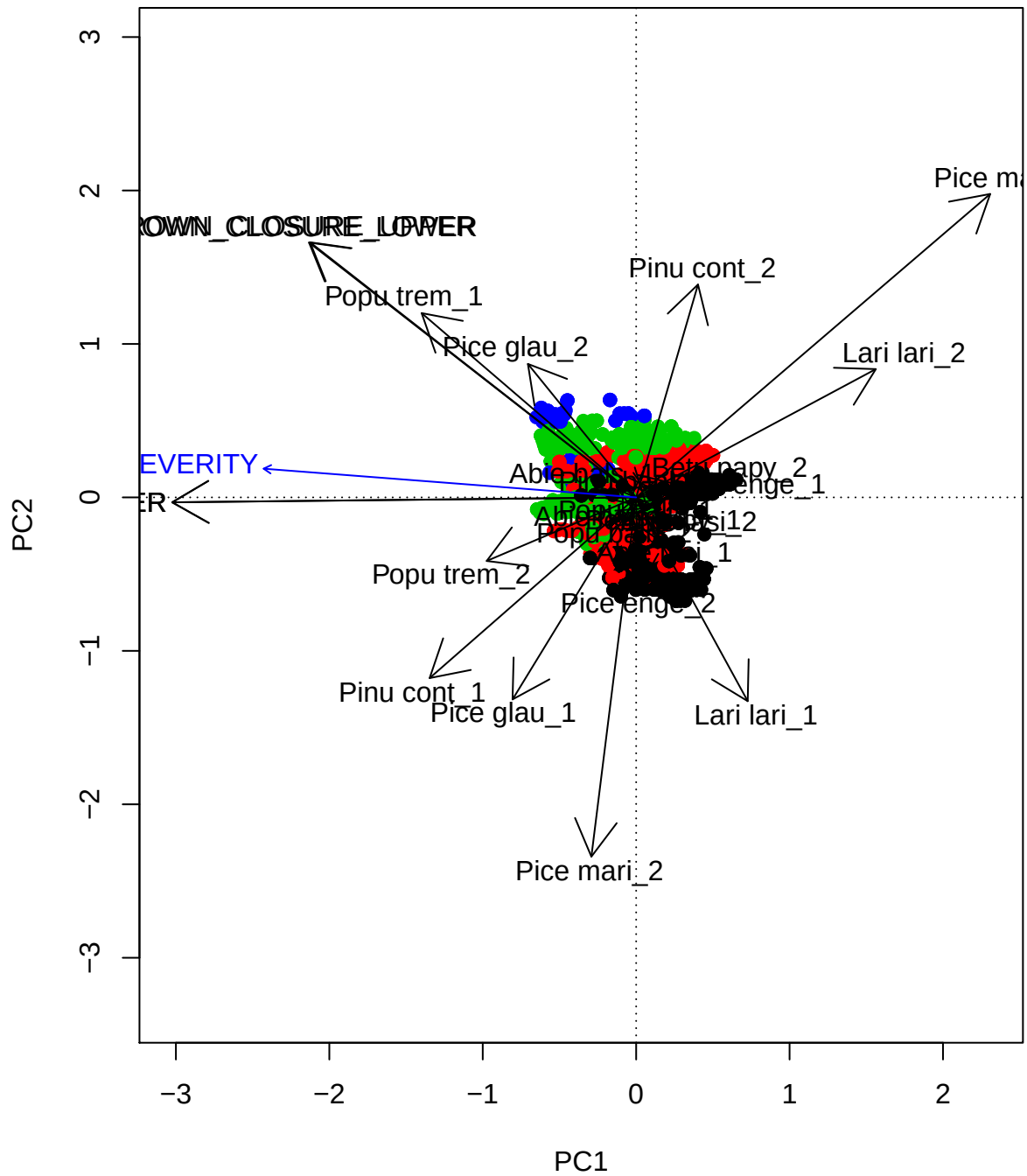


Correlations between third most dominant species and axes are quite low. Results changed quite a bit, with Lodgepole pine and crown closure appearing both more related to severity, than before (and Aspen and White spruce losing importance).



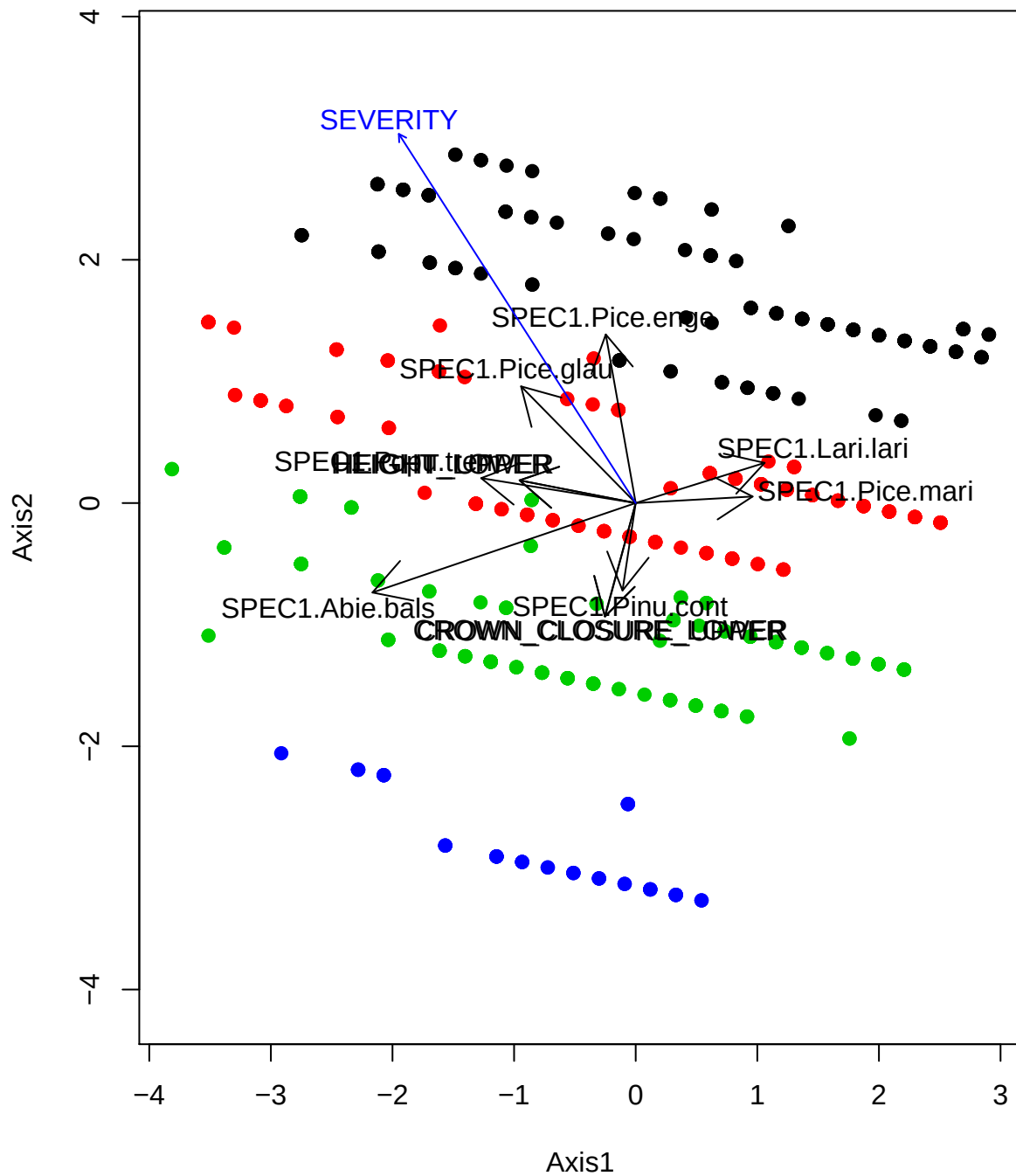
Remove origin - correlations with axes are quite low

Coloured by crown closure



Consider pure stands only collapsing canopy and understory

Coloured by crown closure



SIMPLIFY ANALYSIS/DATA

- Collapse climate data using summary statistics
- Convert species data to long format

COMMUNITY COMPOSITION/STRUCTURE PCAs - simplified data

TODO